ALBUKHARY INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTING & INFORMATICS

Course Title : **OPERATING SYSTEM**
Course Code : **CCC2123**
Tutorial Group : **C**

PROJECT(30%)

A MULTI-PERSPECTIVE COMPARISON OF OPERATING SYSTEMS

GROUP LEADER	**MD PARVEJ AHMED RAFI**	**[AIU23102301]**
GROUP MEMBERS	**Abdullahi Yasin Hersi**	**[AIU21102400]**
Topic	**A Multi-perspective Comparison of Operating System(Windows 10 pro and Kali linux)**	

Table of Content

1. Introduction:

The purpose of this report is to provide a detailed comparative analysis of two distinct operating systems: Windows 10 pro and Kali Linux. Windows 10, developed by Microsoft, is a proprietary operating system widely recognized for its versatility and popularity among personal and professional users. It represents the latest evolution in Microsoft's operating systems, featuring a modernized user interface, extensive integration with other Microsoft services, and an emphasis on productivity and accessibility. Its appeal lies in its ability to cater to a wide range of users, from casual individuals to professionals and gamers.

On the other hand, Kali Linux, developed and maintained by Offensive Security, is a specialized Linux distribution tailored for cybersecurity enthusiasts and professionals. Its primary focus is on penetration testing, digital forensics, and ethical hacking. Unlike general-purpose operating systems, Kali Linux comes preloaded with an extensive suite of tools designed specifically for identifying and resolving security vulnerabilities. This makes it an indispensable resource for IT professionals and organizations focused on securing digital infrastructures.

This report explores the unique characteristics and functionalities of both operating systems, comparing them across several key areas: user interface, system performance, software ecosystem, hardware compatibility, and security mechanisms. By examining these aspects, the report aims to highlight their respective strengths, limitations, and suitability for different use cases. The comparison provides insights into how each system aligns with its target audience and their specific needs, whether it be general productivity, gaming, or cybersecurity and forensic tasks.

Windows 10 emphasizes a seamless and user-friendly experience, offering features such as snap layouts for multitasking, extensive software support, and broad hardware compatibility. It integrates deeply with Microsoft's ecosystem, including Office 365 and cloud services like OneDrive, to enhance productivity and collaboration. With its polished design and robust gaming capabilities powered by DirectX 12, it appeals to both casual users and professionals.

Conversely, Kali Linux is designed with a more technical audience in mind, focusing on efficiency, lightweight performance, and powerful tools for ethical hacking and security testing. It provides users with extensive customization options and supports multiple desktop environments to suit specific preferences. Its open-source nature encourages community-driven development and ensures transparency, making it a trusted platform for critical cybersecurity operations.

2. User Interface (UI) and User Experience (UX):

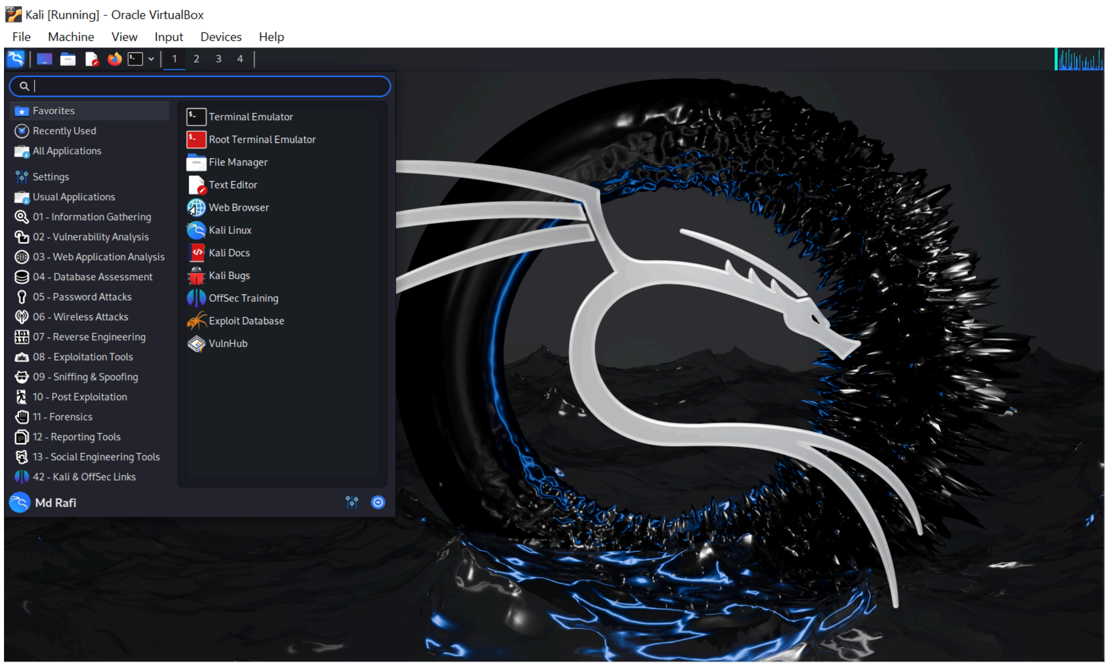
- **Windows 10:**
  Windows 11 features a sleek, modern interface with a centered taskbar, a redesigned Start menu, and "Snap Layouts" for easy multitasking. It offers moderate customization options and focuses on simplicity and integration with Microsoft services, making it user-friendly and efficient for personal and professional use.
- **Kali Linux:**
  Kali Linux uses the lightweight XFCE desktop environment, designed for performance and efficiency. It allows extensive customization but requires technical expertise, making it ideal for advanced users. Its streamlined interface ensures smooth operation, even on older hardware, and supports its suite of cybersecurity tools.

2.1 Desktop Environment:

2.1.1 Windows Explorer (Windows 11):

- Default desktop environment for Windows 11, designed for ease of use and familiarity.
- Features a visually appealing interface with a centered taskbar, a redesigned Start menu, and quick access to frequently used applications.
- Built-in file manager allows efficient navigation and organization of files.
- Limited customization options, focusing on simplicity and productivity.

2.1.2 XFCE (Kali Linux):



- Lightweight and efficient desktop environment, default for Kali Linux.
- Optimized for performance, ensuring smooth operation even on low-resource hardware.
- Highly customizable, allowing users to modify panels, layouts, and themes to suit their preferences.
- Supports alternative desktop environments (e.g., GNOME, KDE) for those who prefer different interfaces.

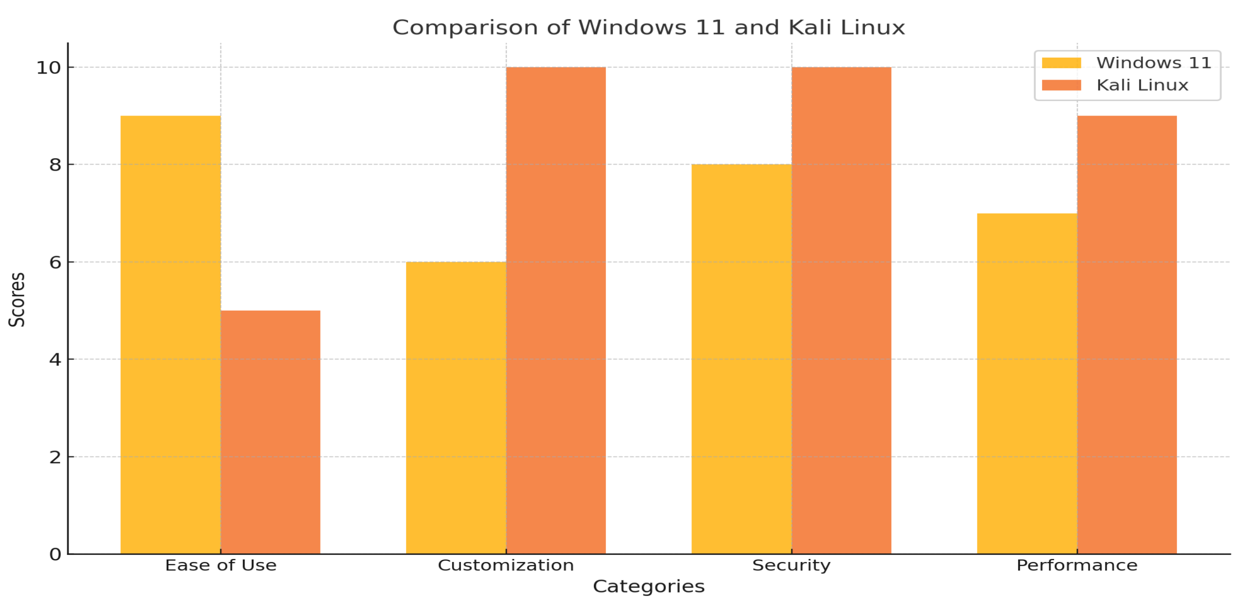
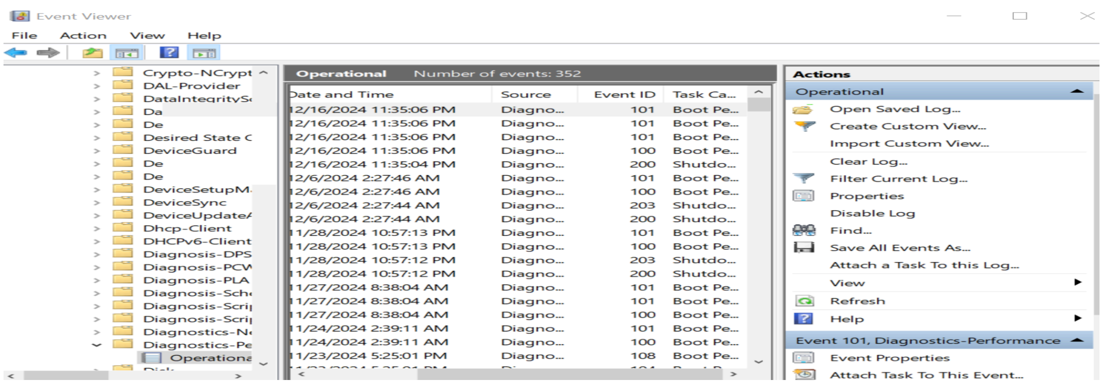


Fig.: comparison of Windows And Kali Linux

3. System Performance and Resource Optimization

3.1 Windows 10 Pro:

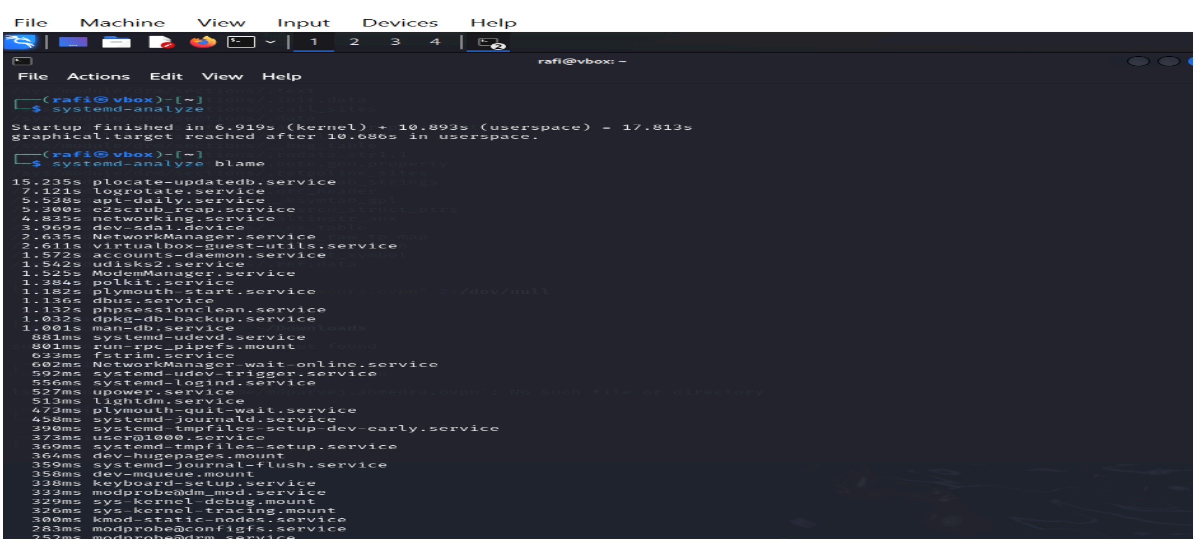
3.1.1 Windows 10 Pro Boot Time:



Windows 11 offers a modern and responsive system, but its boot time is relatively moderate, averaging around 20 seconds on an SSD. However, resource usage can be demanding, particularly during multitasking, as the operating system utilizes higher amounts of CPU and RAM to support its polished interface and integrated features. This makes it suitable for systems with robust hardware but may strain older or low-spec devices.

3.2 Kali Linux:

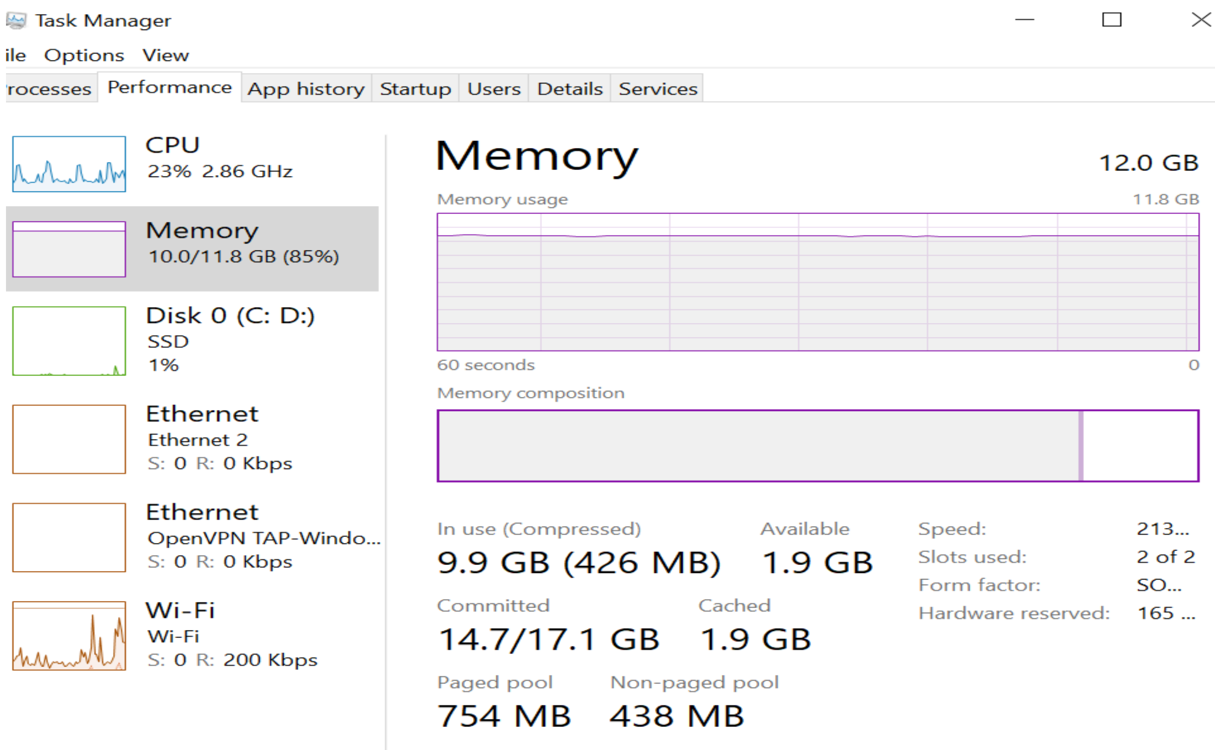
3.2.1 Kali Linux Boot Time:



Kali Linux boots faster, with a time ranging from **10 to 15 seconds**. This efficiency is attributed to its lightweight XFCE desktop environment and streamlined architecture, emphasizing speed and performance, especially for users prioritizing functionality in cybersecurity tasks.

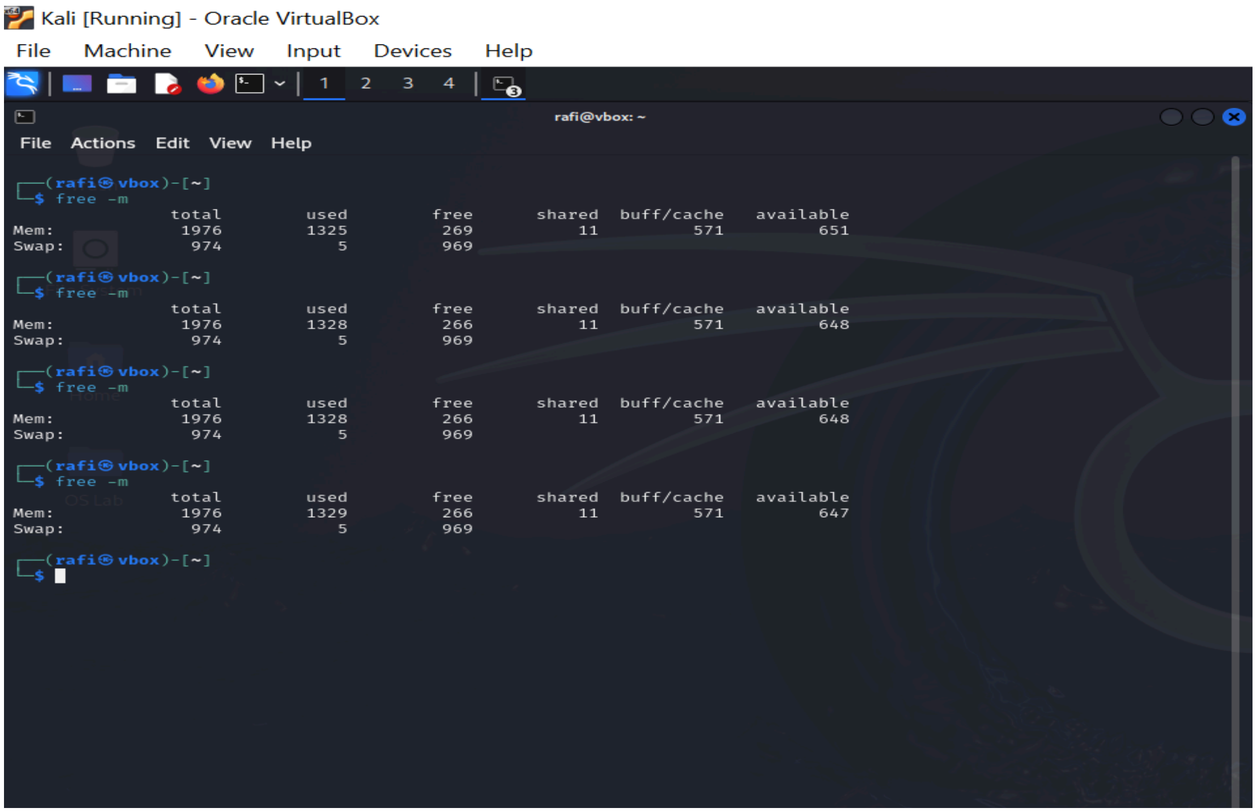
3.3 Memory Usage:

3.3.1 Windows 10 Pro Memory Usage (Idle):



Windows 11 utilizes approximately **2.5 GB of RAM** when idle. This higher memory usage reflects the modern and feature-rich design of the operating system, which includes a visually appealing interface, background processes, and integrated services tailored for a seamless user experience.

3.3.2 Kali Linux Memory Usage (Idle):



Kali Linux, on the other hand, consumes around **1 GB of RAM** in an idle state. Its lower memory usage is due to its lightweight XFCE desktop environment and streamlined focus on performance, making it an efficient choice for resource-conscious users and specialized tasks like penetration testing.

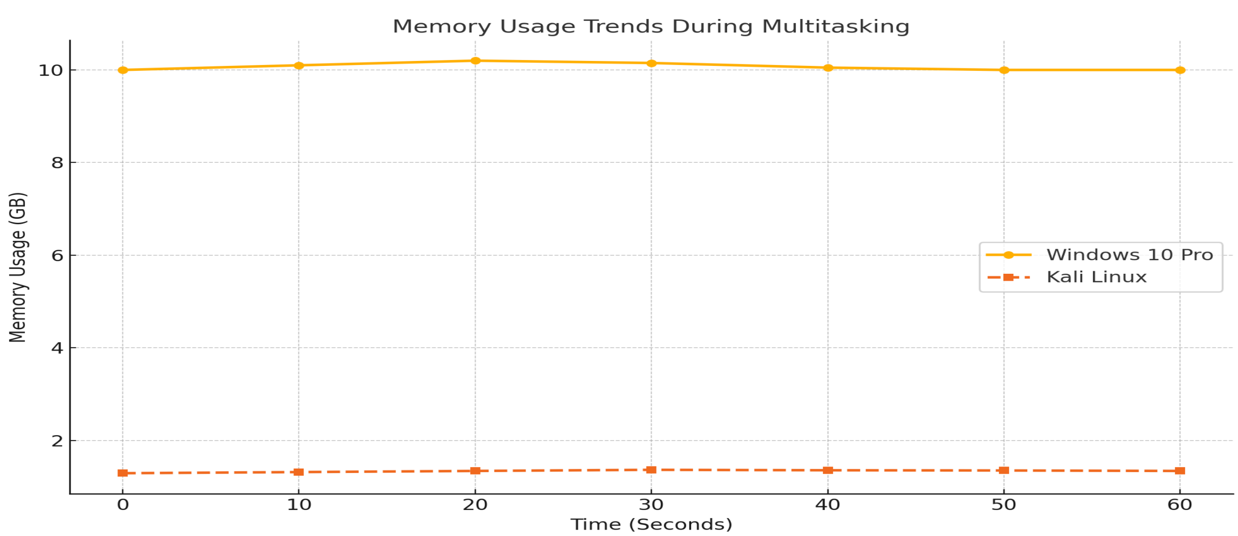


Fig: Memory Usage Trends During Multitasking

4. Software and Development Ecosystem:

4.1 Software Availability:

Windows 10 Pro offers an extensive library of software suitable for both personal and professional use. Its strong support for gaming platforms like Steam and Epic Games, along with enterprise-grade tools, makes it a versatile choice for a wide range of users. On the other hand, Kali Linux comes pre-installed with a comprehensive suite of cybersecurity tools such as Metasploit, Wireshark, and Nmap. This makes Kali Linux a specialized platform for professionals in penetration testing, network security, and digital forensics.

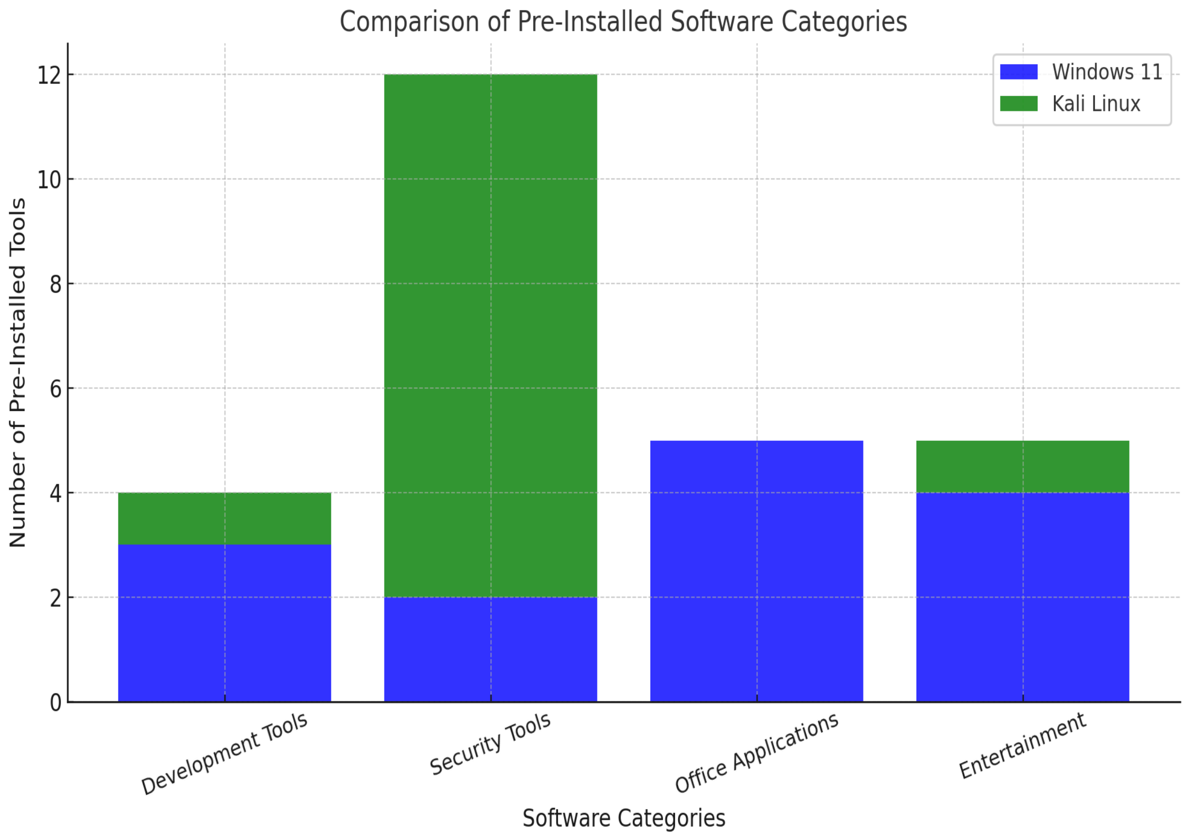


Fig : kali linux and windows 10 pro software Categories

4.2 Development Tools Comparison:

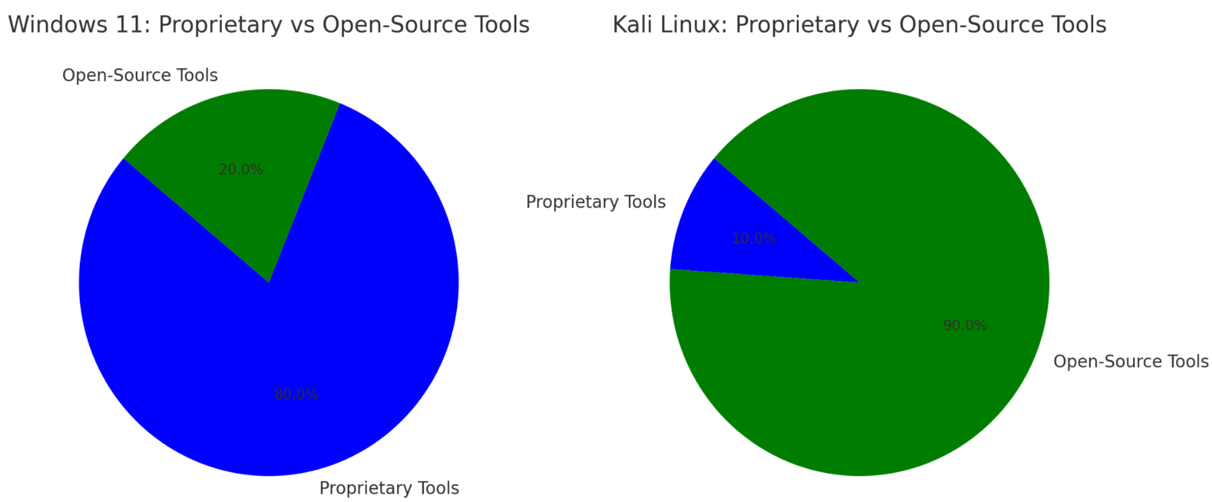


Fig: kali linux and windows 10 pro Development tools

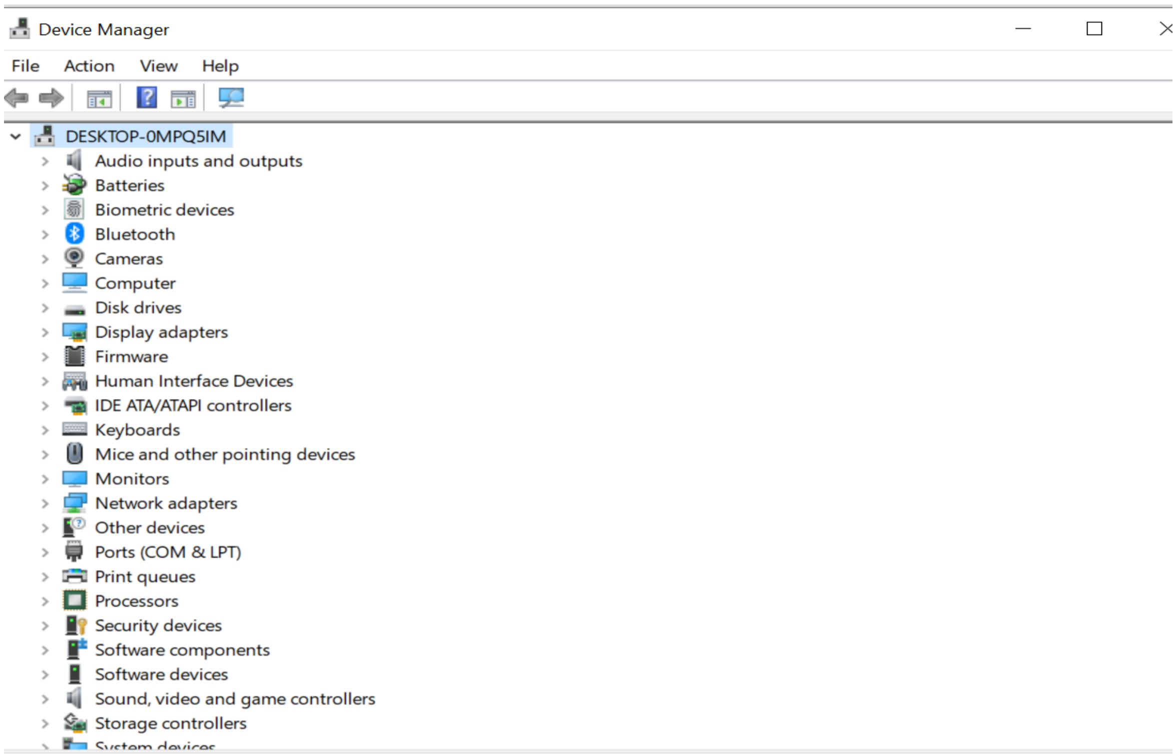
Integrated Development Environments (IDEs):
Windows 10 Pro supports a wide range of IDEs, including Visual Studio, IntelliJ IDEA, and Eclipse, catering to enterprise-grade development and diverse coding projects. Kali Linux, in contrast, offers IDEs like VS Code, PyCharm, and Eclipse, which emphasize open-source tools and Linux-first compatibility.

Libraries and Tools:
Windows 10 Pro provides a mix of proprietary and open-source libraries, giving developers flexibility and access to Microsoft's ecosystem. Kali Linux, however, focuses primarily on open-source, Linux-native libraries and tools, ideal for security professionals and developers seeking greater freedom in customization.

Integration:
Windows 10 Pro integrates seamlessly with Microsoft services like Office 365, Azure, and Active Directory, making it an excellent choice for businesses and power users. Kali Linux, tailored for security specialists, offers tools for web application testing, network penetration testing, and forensic analysis, prioritizing specialized functionality over general integration.

5. Hardware Compatibility and Optimization:

5.1 Windows 10 Pro Device Support:

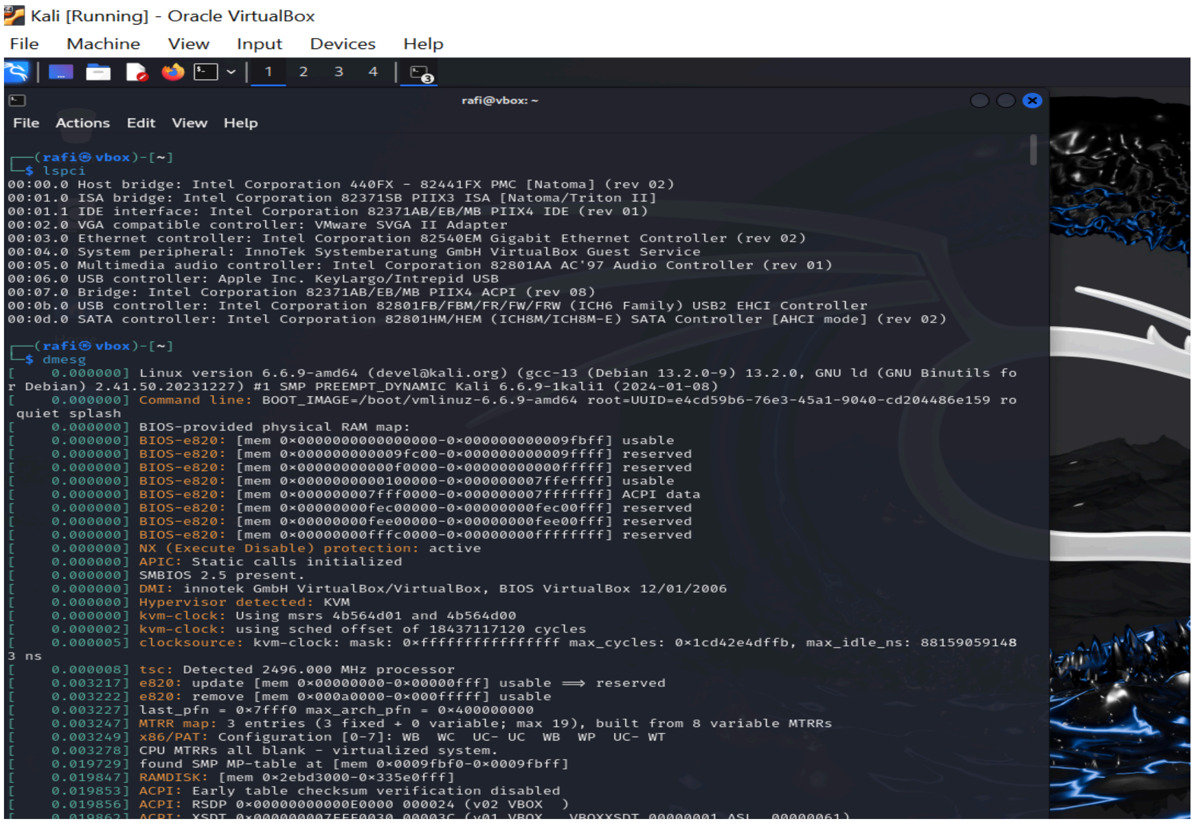


Windows 10 Pro offers broad driver availability and plug-and-play support for most devices. Its extensive ecosystem ensures that peripherals like printers, GPUs, and IoT devices work seamlessly out of the box, making it an excellent choice for users who prioritize ease of use and quick setup.

Compatibility Metrics for Windows 10 Pro:

- Printer Support: Windows 10 Pro provides reliable plug-and-play compatibility, ensuring printers can be easily connected without the need for extensive configuration.
- GPU Compatibility: High compatibility with major GPU brands such as NVIDIA and AMD ensures smooth performance for gaming, design, and computational tasks.
- IoT Devices: Offers moderate support for IoT devices, with consistent improvements through regular updates to accommodate smart home and enterprise IoT solutions.

5.2 Kali Linux Device Support:



Kali Linux supports a wide range of devices, but manual driver installation may be required for certain peripherals. This is especially true for less commonly used hardware, making it better suited for experienced users comfortable with Linux system configuration.

Compatibility Metrics for Kali Linux:

- Printer Support: Printer setup in Kali Linux often requires manual configuration, with varying levels of difficulty depending on the printer model and driver availability.
- GPU Compatibility: While compatible with major GPU brands, Kali Linux often needs manual setup and configuration to ensure optimal performance, particularly for advanced GPU features.
- IoT Devices: Limited to moderate support for IoT devices, with a focus on security testing and research rather than everyday usability.

6. Security and Privacy Mechanisms:

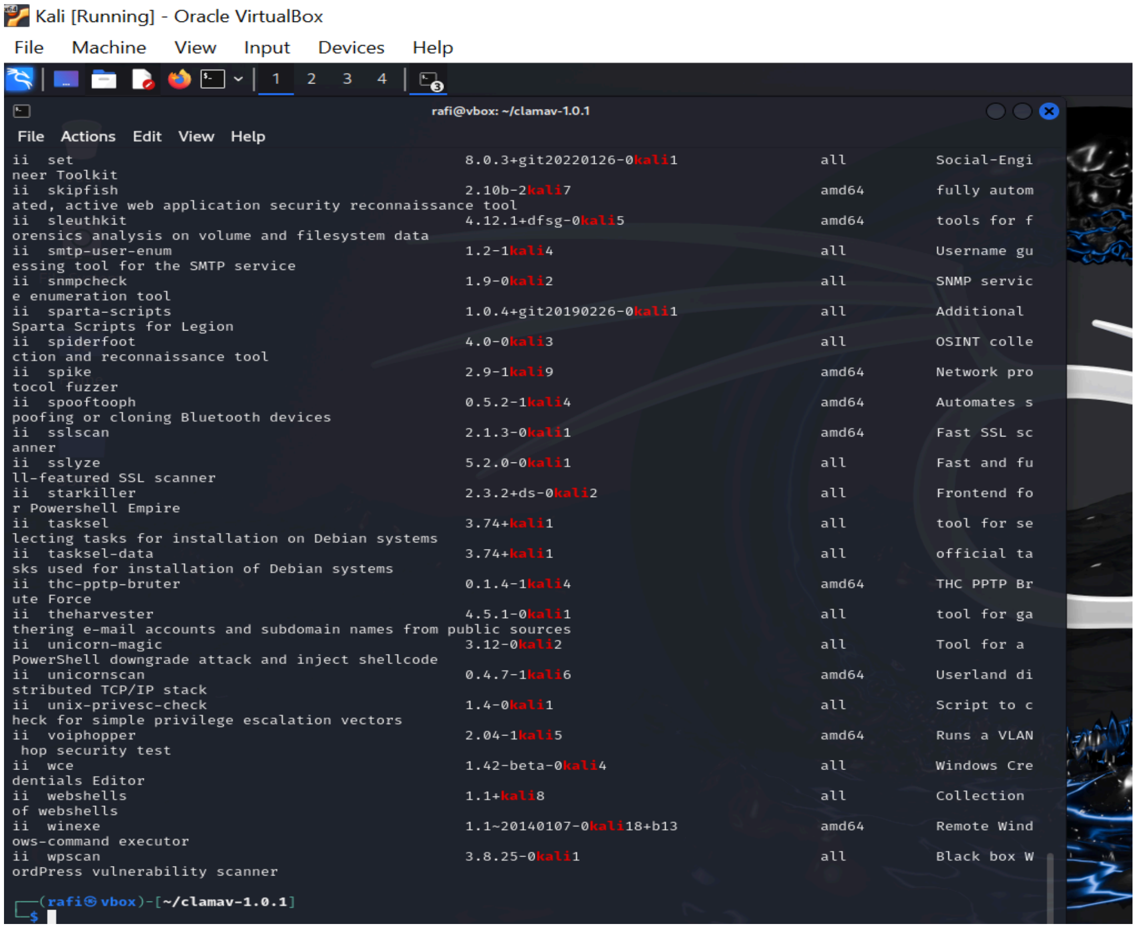
6.1 Windows 10 Pro Security Features:

Windows 10 Pro offers a comprehensive suite of security features designed to protect users from malware, secure their data, and provide privacy options.

- Malware Protection: Windows 10 Pro includes Windows Defender, a built-in antivirus and anti-malware solution that provides real-time protection, regular updates, and system-wide scanning capabilities.
- Encryption: BitLocker encryption is available in Windows 10 Pro, enabling users to secure their drives and protect sensitive data with advanced encryption algorithms.
- Privacy: Windows 10 Pro comes with data collection enabled by default, primarily for diagnostics and service improvements. Users can adjust privacy settings through the Windows Settings app to limit data sharing and improve privacy.

Windows 10 Pro's security features are user-friendly and designed for both personal and professional environments, offering a balance between usability and robust protection.

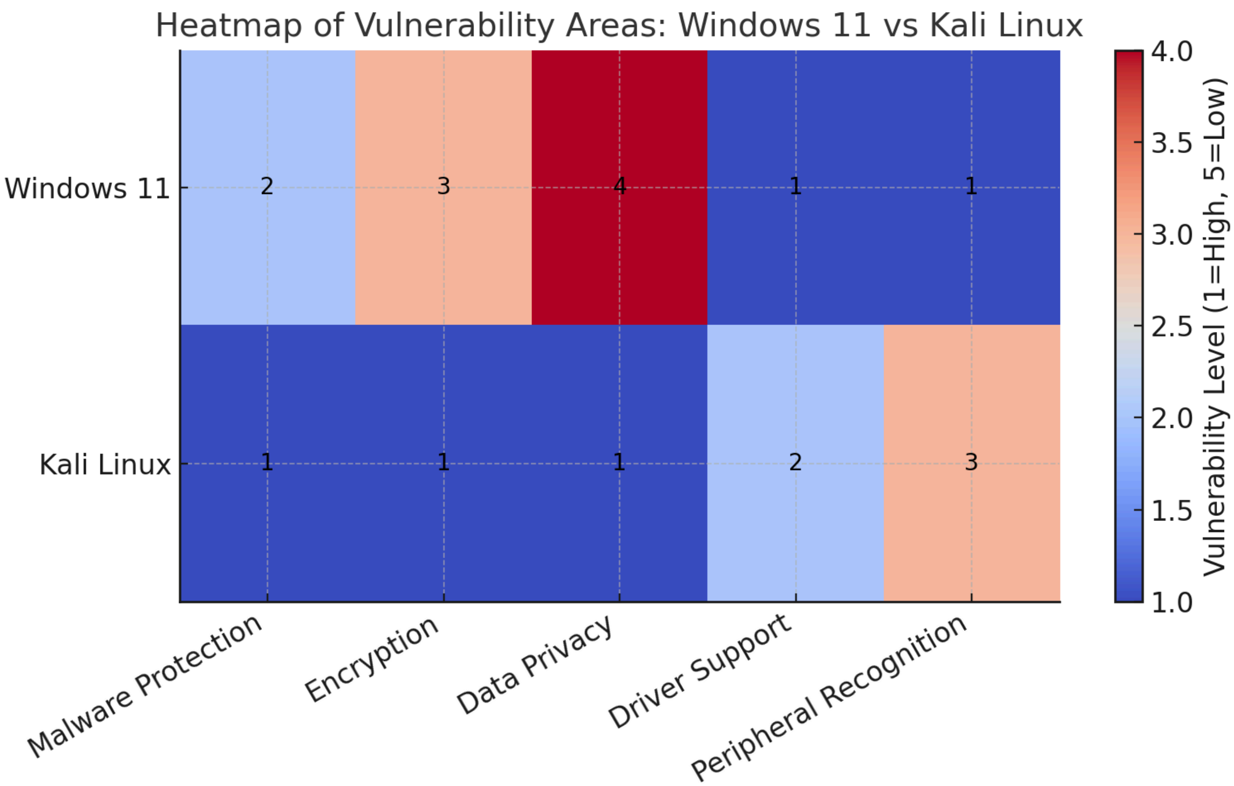
6.2 Kali Linux Security Features:



Kali Linux focuses on security and privacy, with features tailored for professionals working in penetration testing, cybersecurity, and forensics.

- Malware Protection: Kali Linux supports tools like ClamAV for detecting and mitigating malware threats. While not pre-installed, these tools are easily added and configured.
- Encryption: Native support for LUKS (Linux Unified Key Setup) ensures strong encryption for securing data on drives, providing a reliable method for protecting sensitive information.
- Privacy: Kali Linux adheres to open-source principles, implementing minimal data collection by default. This focus on privacy ensures users retain full control over their data and system operations without needing extensive adjustments.

Kali Linux's security features are designed for advanced users who require granular control and high levels of privacy, making it a go-to choice for security professionals.



config :Heatmap of vulnerability Areas: Windows 11 vs. Kali Linux

7. Conclusion:

In summary, Windows 10 Pro and Kali Linux cater to distinct user needs, making each operating system uniquely suited to specific use cases. Windows 10 Pro excels in user-friendliness, broad device compatibility, and an extensive ecosystem, offering features such as plug-and-play device support, seamless integration with Microsoft's tools, and robust built-in security with Windows Defender and BitLocker encryption. It is a strong choice for both personal and professional users seeking convenience, compatibility, and productivity.

On the other hand, Kali Linux focuses on performance, flexibility, and privacy, making it an ideal choice for advanced users, particularly cybersecurity professionals. Its pre-installed suite of penetration testing tools, open-source architecture, minimal data collection, and native LUKS encryption highlight its specialized nature. While it requires more technical expertise for setup and device management, Kali Linux delivers unparalleled capabilities in security and forensic analysis.

Ultimately, the choice between Windows 10 Pro and Kali Linux depends on user priorities. For general users and businesses, Windows 10 Pro provides a comprehensive and accessible environment. For security experts and Linux enthusiasts, Kali Linux offers a highly specialized and privacy-focused platform. Both systems stand out in their respective domains, serving as valuable tools for their target audiences.

8. References:

1. Awan, M. T., & Khan, K. (2022). Linux vs. Windows: A Comparison of Two Widely Used Platforms. *Journal of Computer Science and Technology Studies*, *4*(1), 41-53. https://www.al-kindipublisher.com/index.php/jcsts/article/download/2763/2420
2. Softić, J., & Vejzović, Z. (2022, March). Windows 10 Operating System: Vulnerability Assessment and Exploitation. In *2022 21st International Symposium INFOTEH-JAHORINA (INFOTEH)* (pp. 1-5). IEEE. https://ieeexplore.ieee.org/abstract/document/9751274/
3. Ghafarian, A. (2019). Using kali linux security tools to create laboratory projects for cybersecurity education. In *Proceedings of the Future Technologies Conference (FTC) 2018: Volume 2* (pp. 358-367). Springer International Publishing.https://link.springer.com/chapter/10.1007/978-3-030-02683-7_25
4. Yosifovich, P., Ionescu, A., Russinovich, M. E., & Solomon, D. A. (2017). *Windows Internals: System architecture, processes, threads, memory management, and more, Part 1*. Microsoft Press.https://books.google.com/books?hl=en&lr=&id=y83LDgAAQBAJ&oi=fnd&pg=PA1999&dq=memory+storage+windows&ots=kJ9uSCLsjz&sig=wcStfyNGcs_9SNHqol_AnfIj65E
5. Hertzog, R., O'Gorman, J., & Aharoni, M. (2017). Kali linux revealed. *Mastering the Penetration Testing Distribution*.https://www.academia.edu/download/63524487/ch120200604-27390-1kbkd1l.pdf

9. VIDEO PRESENTATION:

https://drive.google.com/file/d/1vkuSfNTVBtSwQDO6QV-WOToT86Grples/view?usp=drive_link